

Smart Deeds for Mastering Digital Electronics

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In this article, we introduce you to Digital Electronics Education and Design Suite, also known as Deeds software, which is a smart learning environment for digital electronics. If we take the conventional Indian education scenario, most of our fresh electronic graduates are not

industry-ready. As technology advances at a rapid pace and the syllabus is not revised at par with it, there is a wide gap between commercial implementation and education and training. We cannot blame the academicians in this regard. If we replace the current syllabus completely with latest technologies, students will be just beating around the bush without

having an understanding of the core fundamental concepts.

Deeds bridges this gap via its learning-by-doing approach. It imparts to students a thorough theoretical foundation on the principles of digital design as well as analysis-cum-problem-solving capabilities and practical-synthesis-cum-design skills to make them industry-ready.

Deeds has been developed by Department of Electrical, Electronic, Telecommunications Engineering and Naval Architecture (DITEN), University of Genoa, Italy. The source code, written in Delphi, now amounts to about 250,000+ lines of code. The latest release of the software, version 1.92.200, is included in this month's EFY Plus DVD.

As a learning environment for logic design

Deeds is designed to be a complete learning environment for digital electronics, covering all major areas of logic design. Students can experience combinational logic networks of complexities varying from simple gates to encoders, multiplexers and so on. They can form a sequential network as simple as a flip-flop to more complex registers and counters. Students can even design finite state machines and do microprocessor based programming.

Freshman-level students can easily understand the low-level behaviour of embedded systems, which is a good foundation for successive courses on design languages, IP cores and hardware-software co-designing.

Latest versions of Deeds come with field programmable gate array (FPGA) support. You do not have to be a master of any hardware-description language (HDL) or high-level

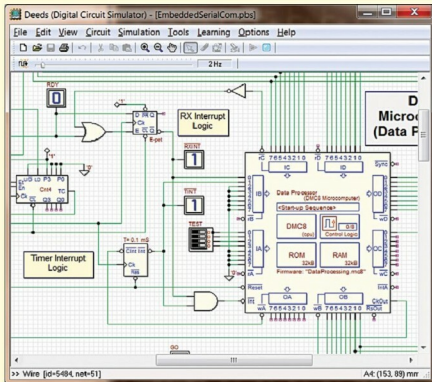


Fig. 1: A circuit based on a microcomputer component in Deeds-DcS

FPGA configuration

Introduction of FPGAs in a first-year course of digital design is now advisable. It is not easy to build a good foundation on logic design just by completely migrating the traditional schematic, simulation and bread-board based prototyping to VHDL based FPGA design flow.

Deeds makes the process of FPGA configuration straightforward and compatible with beginners' skills. Tutorials guide students in the process and, at the same time, provide a synthetic introduction to FPGA projects.

—GiulianoDonzellini, main developer of Deeds



Giuliano Donzellini